

Ethanol Blending in Gasoline – Netherlands

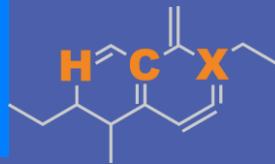
September 2025



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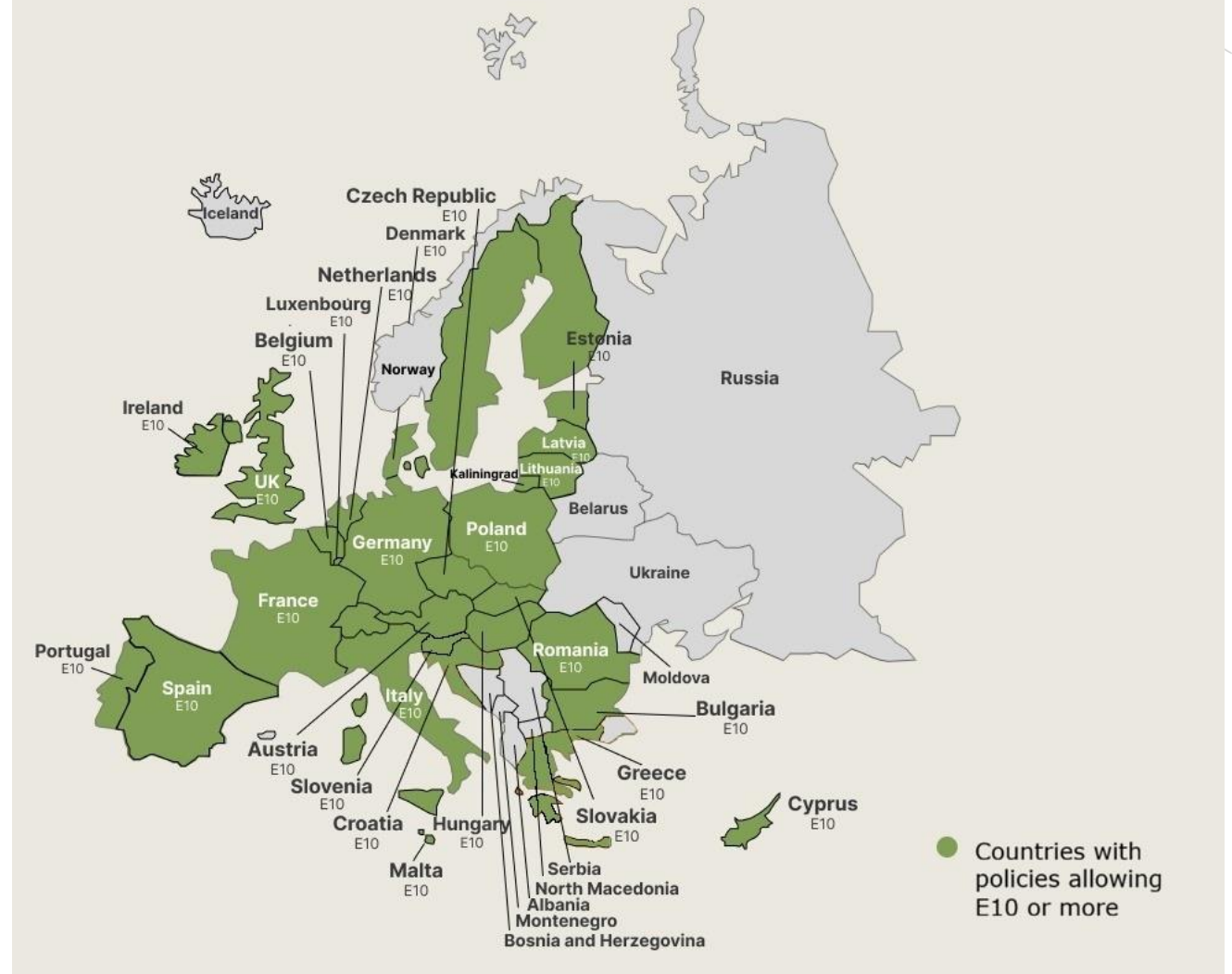


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Netherlands

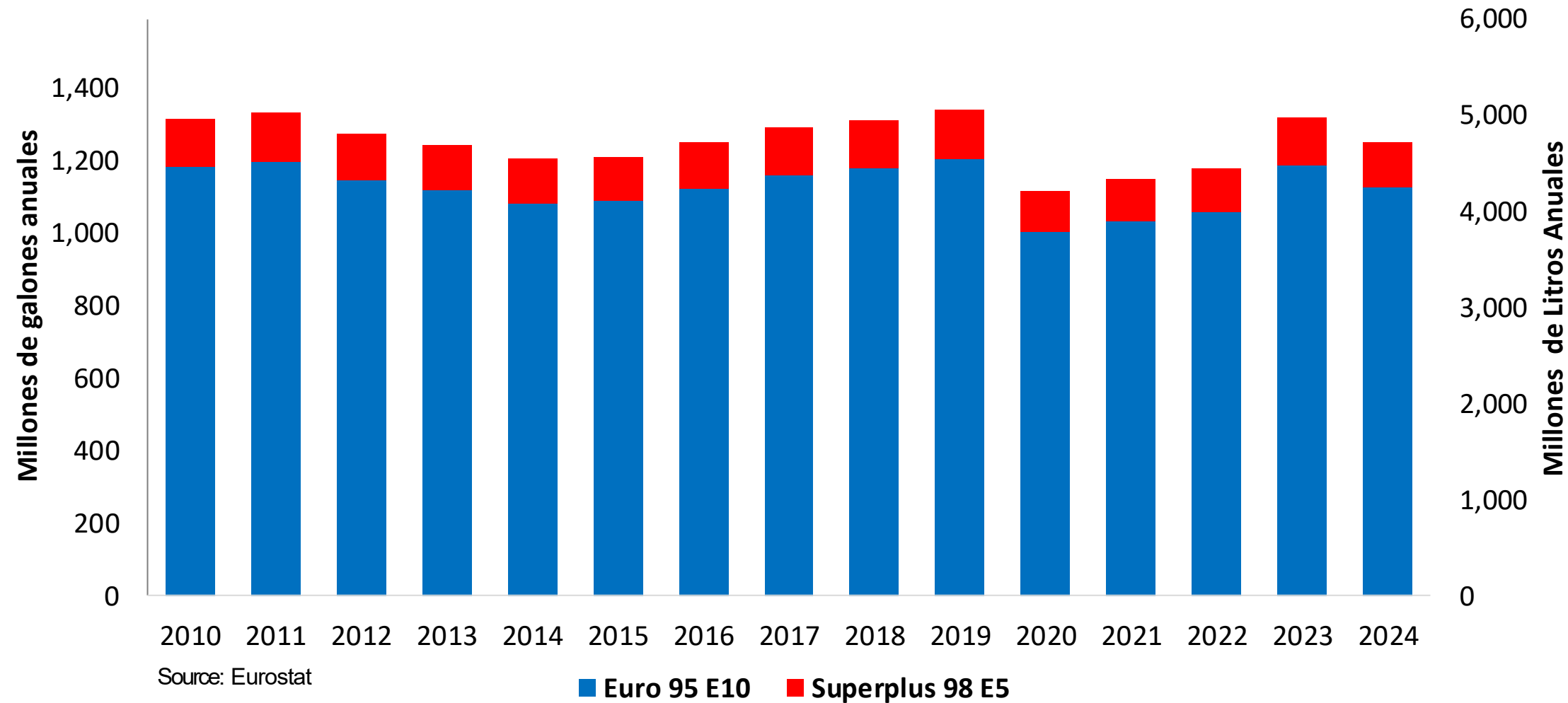


In 2024, gasoline consumption in Netherlands was 4,723 million liters (1,248 million gallons) and growth rate was -1.3%. Gasoline production and net imports were 6,295 and -14,373 million liters (1,663 and -3,797 million gallons) correspondingly. Netherlands complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Netherlands was 507 million liters (134 million gallons) representing 10.7% of gasoline demand. Ethanol production and Net Imports were 1,080 and -574 million liters (285 and -152 million gallons) correspondingly. Ethanol sources are Sugar Cane 5% and Cereals 95%.

Source: Eurostat, European Commission

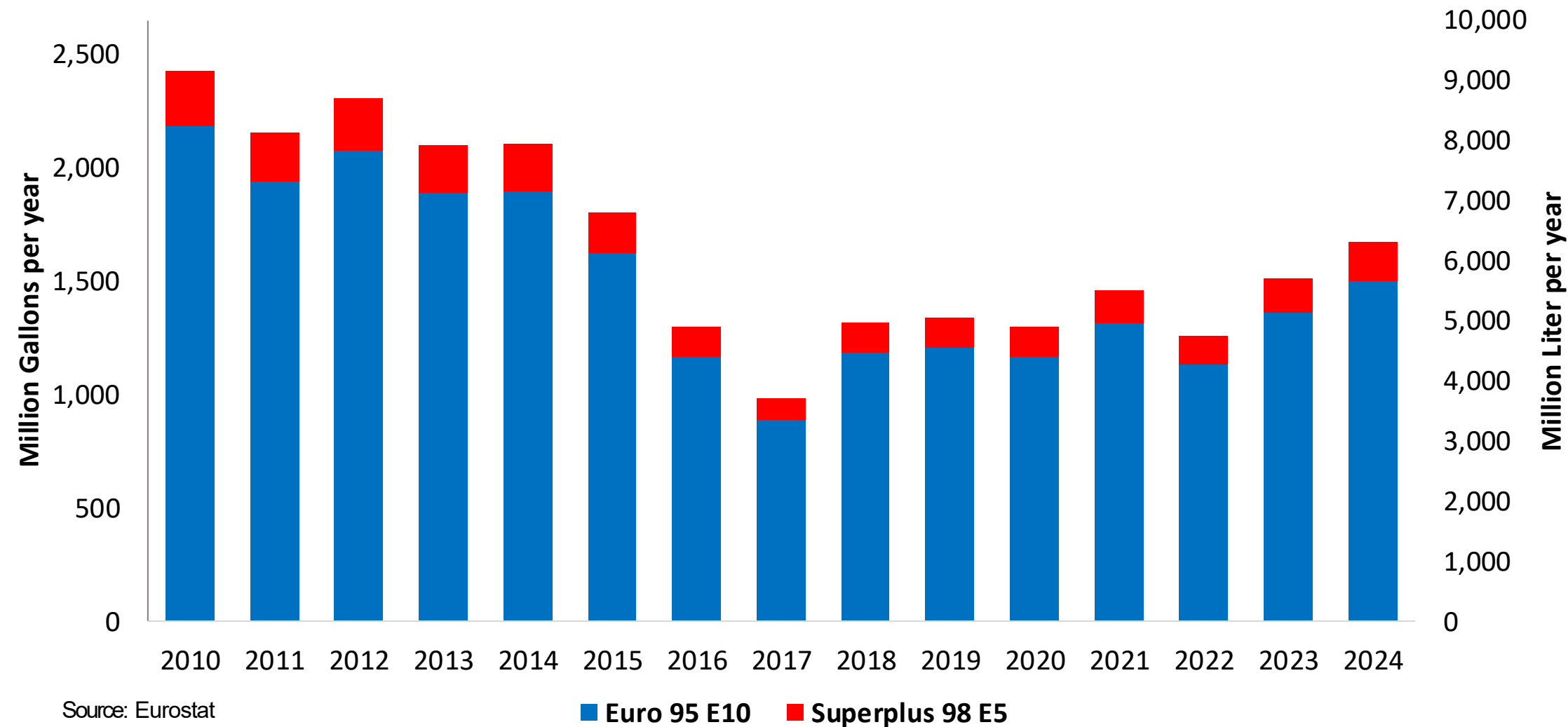
Gasoline Demand by Grade in Netherlands



Gasoline Production in Netherlands

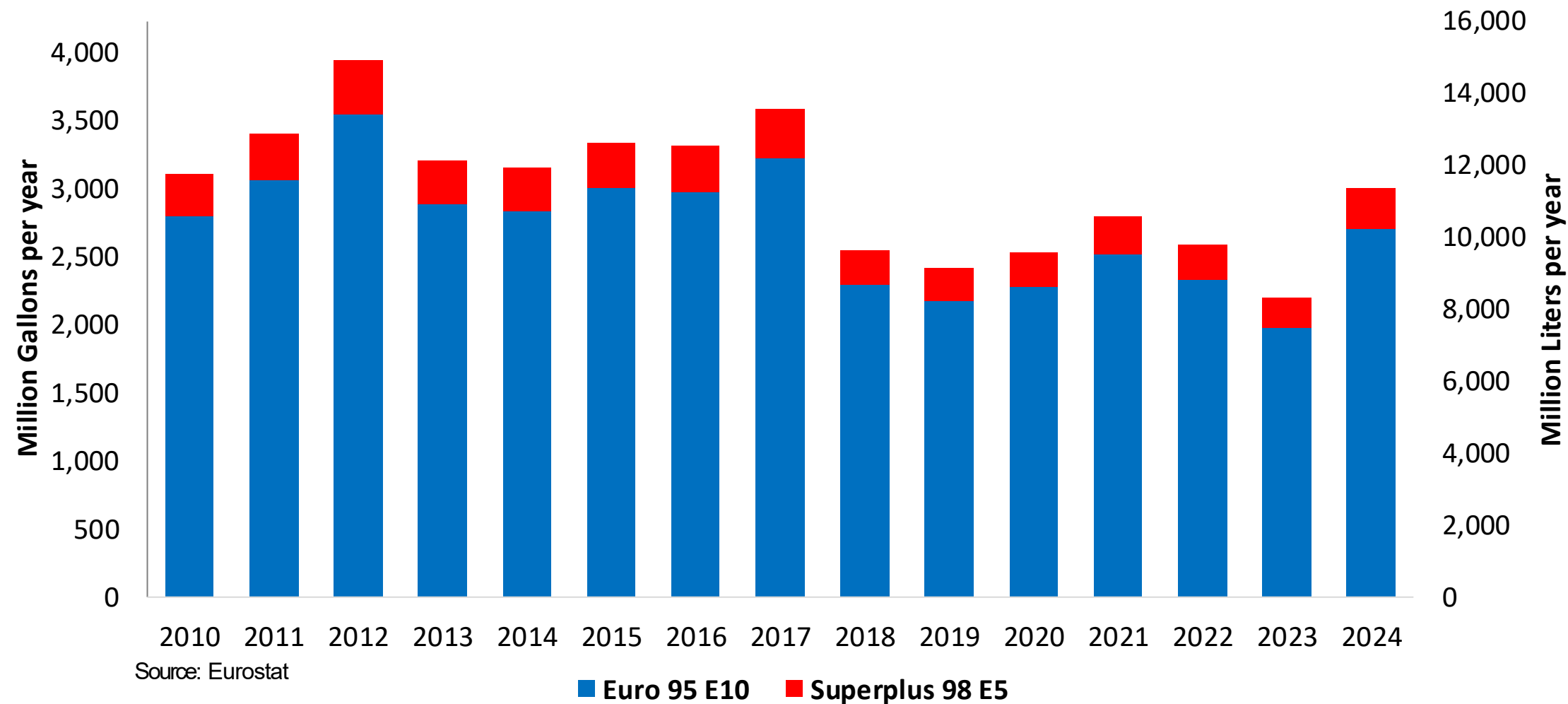


Gasoline Production by Grade in Netherlands

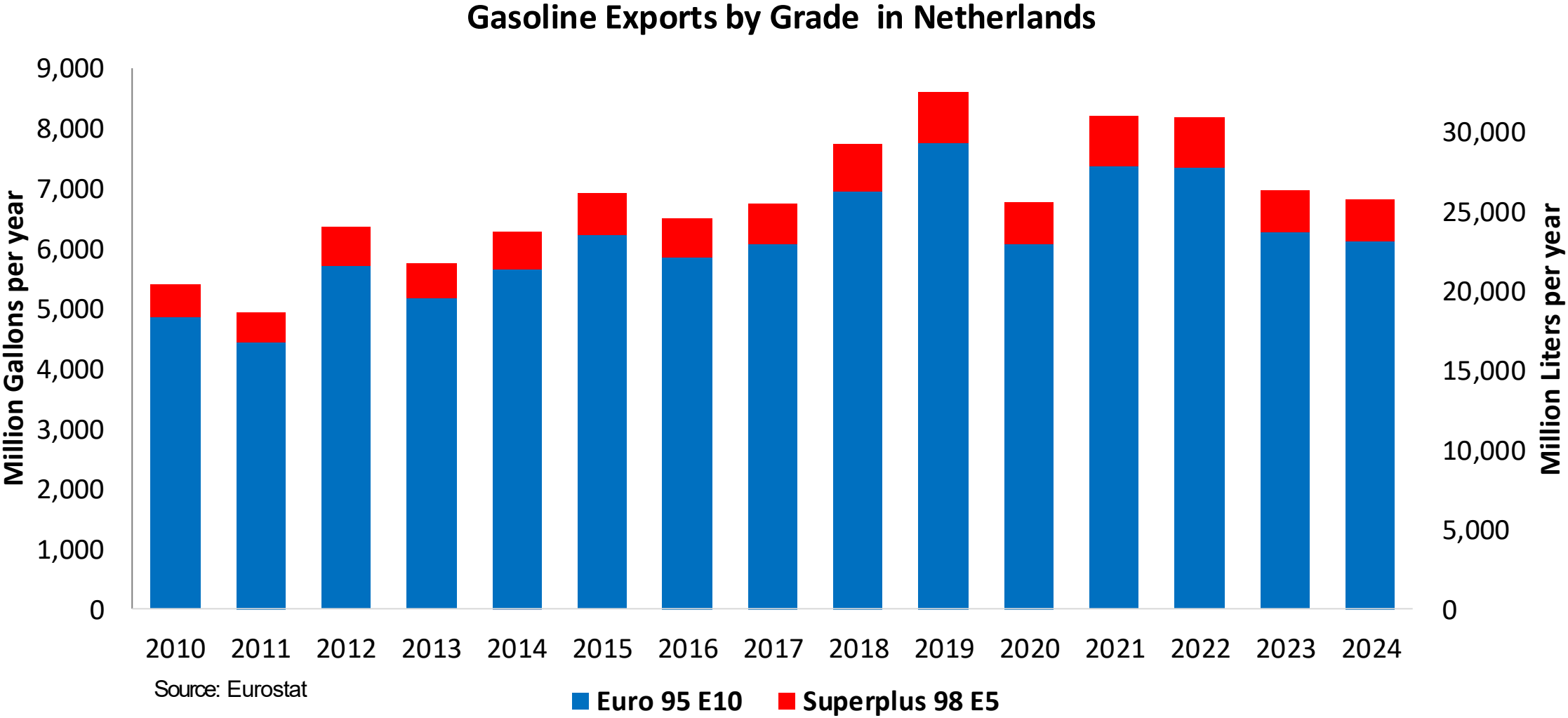


Gasoline Imports in Netherlands

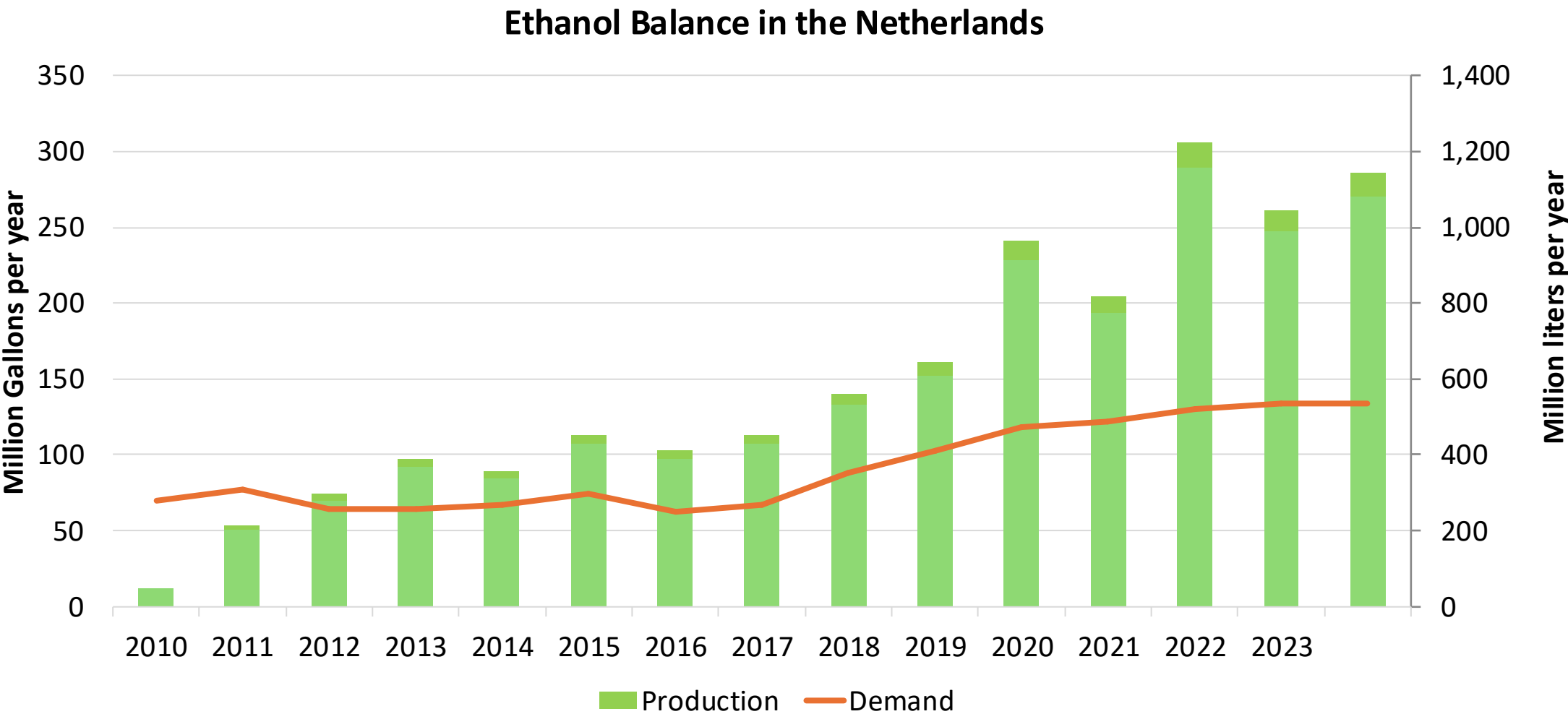
Gasoline Imports by Grade in Netherlands



Gasoline Exports in Netherlands



Ethanol Balance in Netherlands



Source: Eurostat, Statistics Netherlands

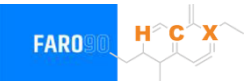
Gasoline Quality in Netherlands

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	28.4							
Advanced Biofuels (%cal)	3.6							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

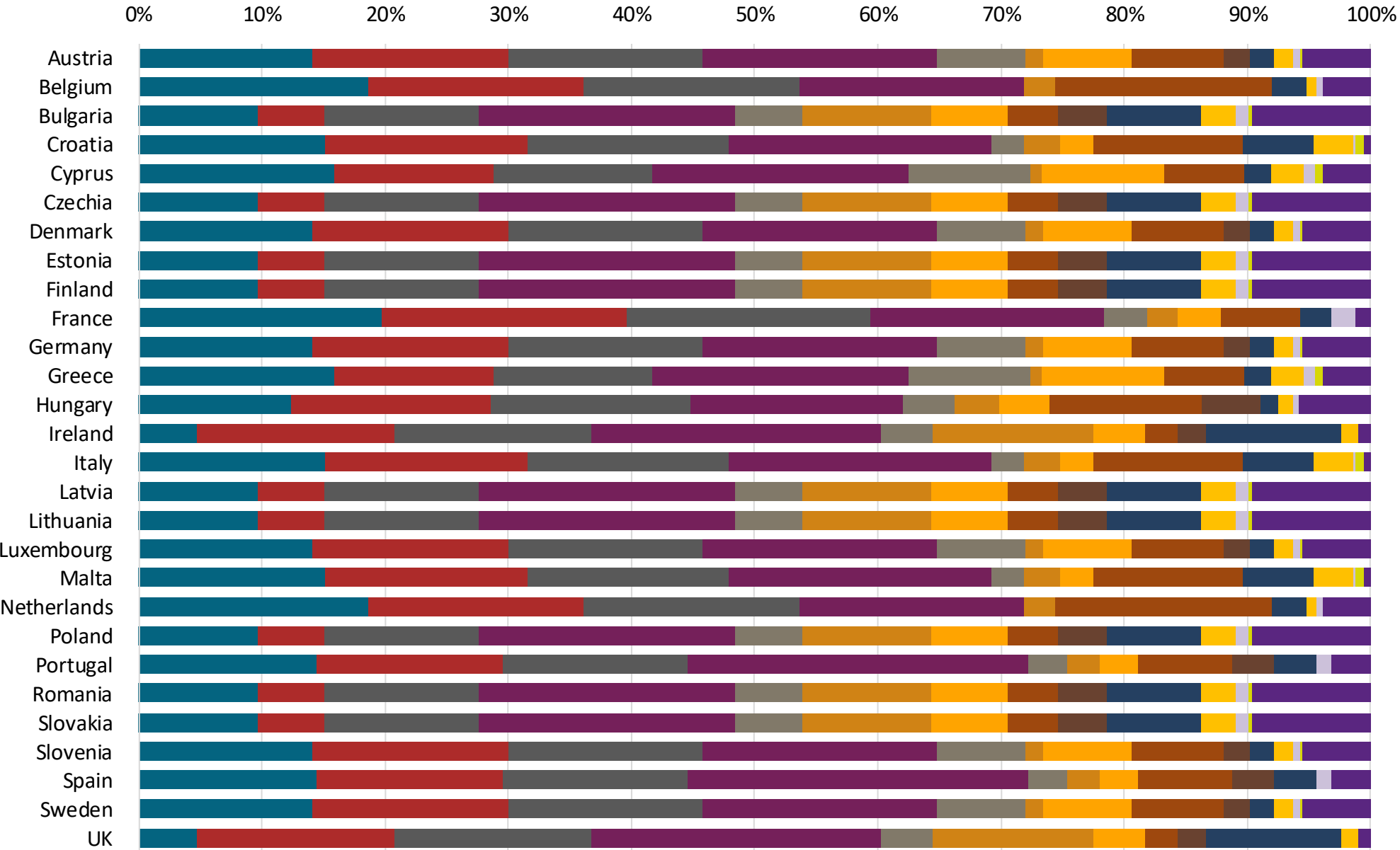
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



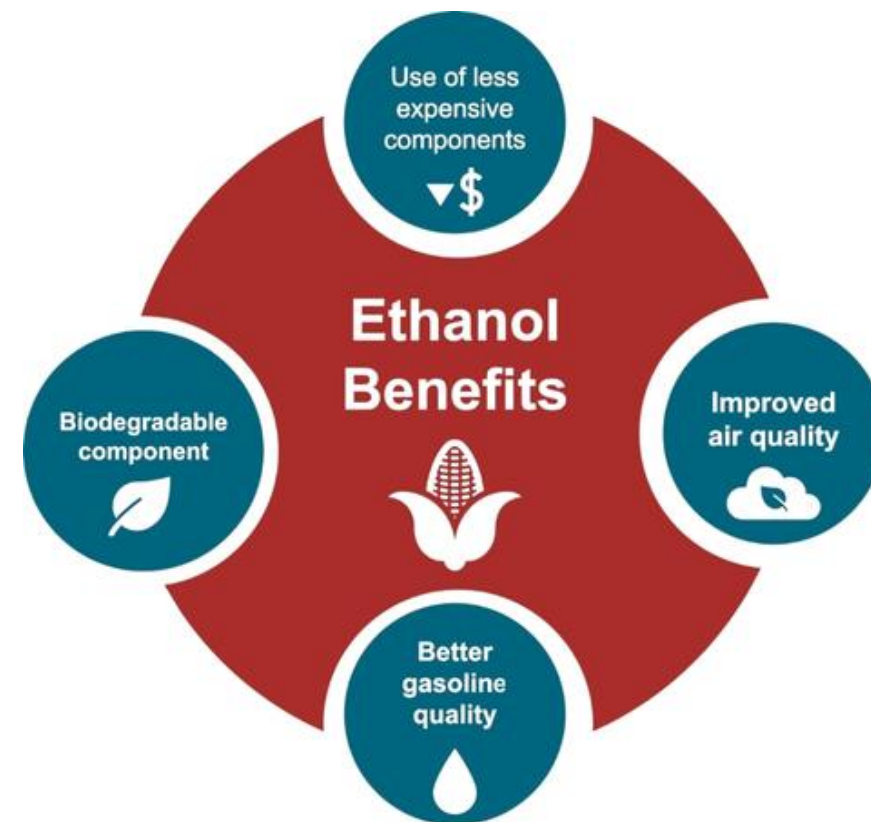
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

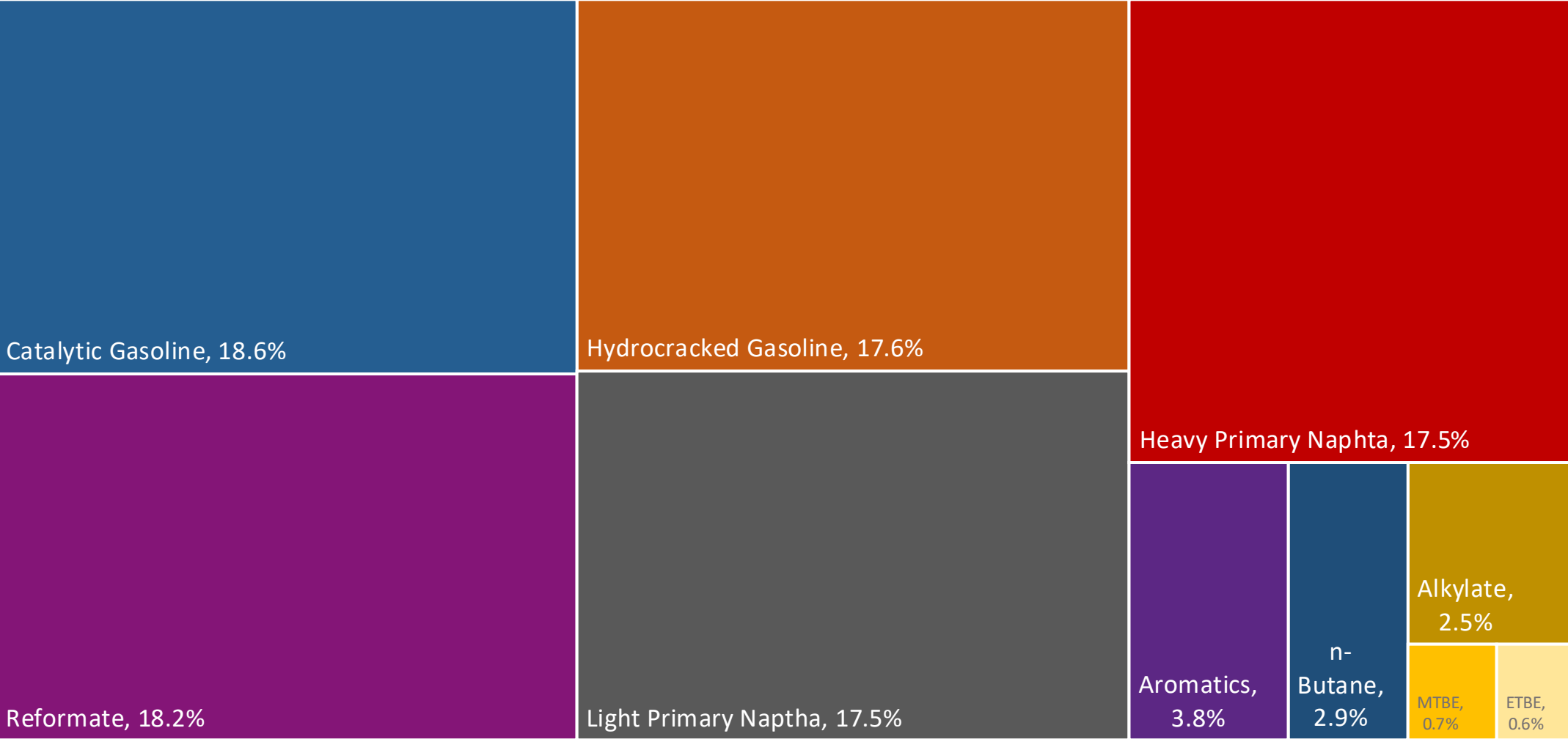
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

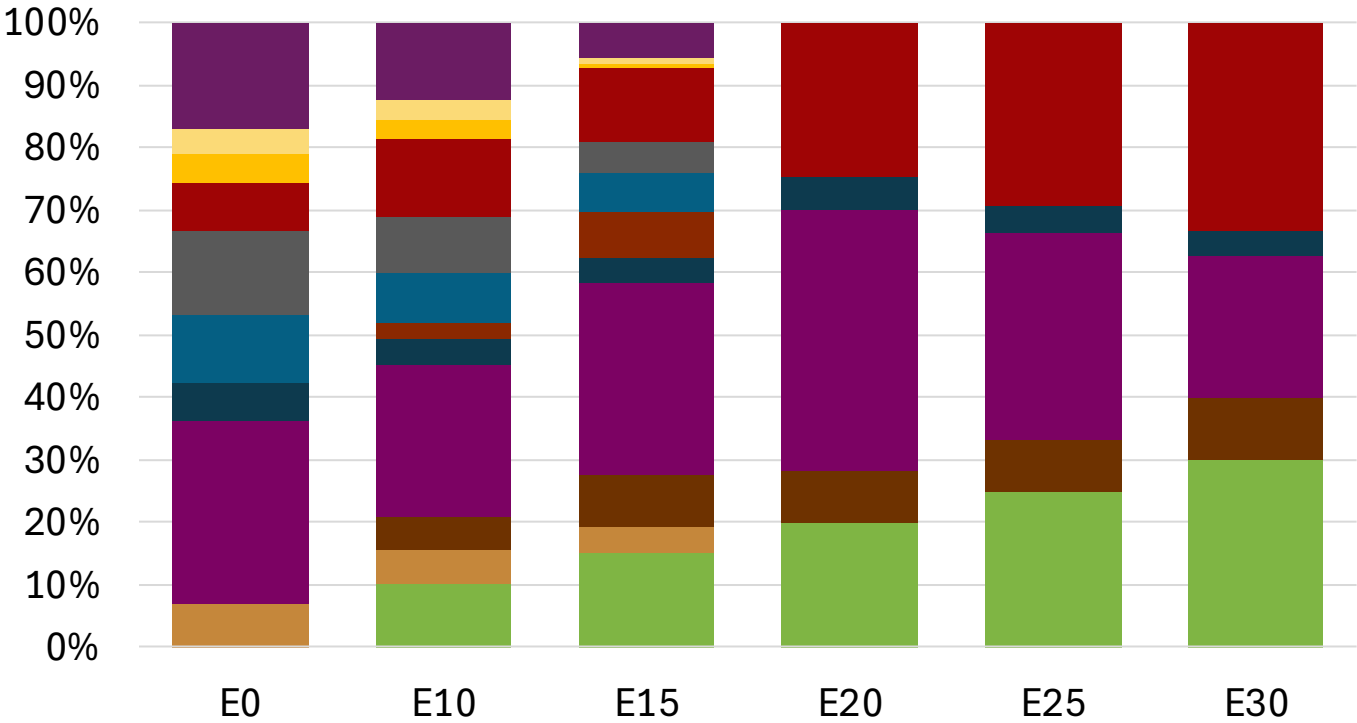
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Netherlands Available Blending Components



Netherlands – Gasoline 95 – Constant Octane

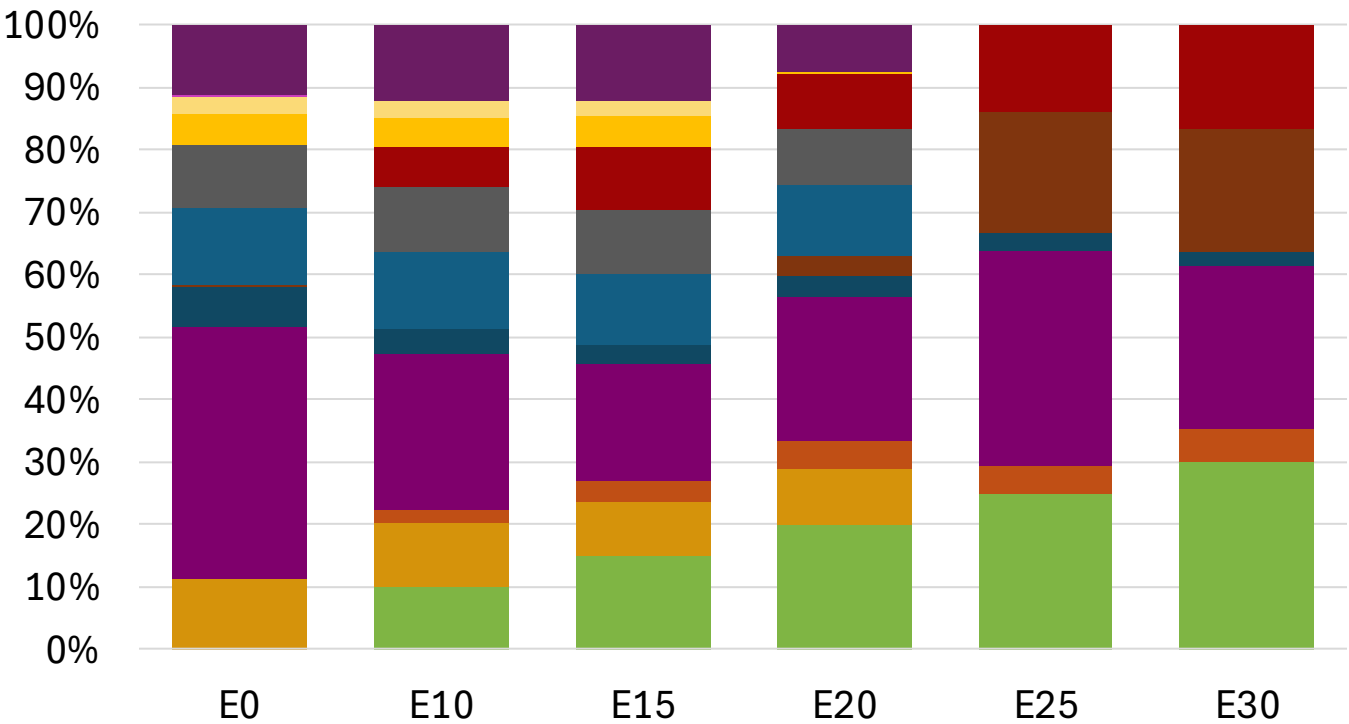
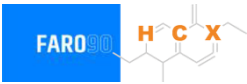


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US\$/gal)	\$ 2.281	\$ 2.147	\$ 2.072	\$ 1.951	\$ 1.875	\$ 1.780

Source: Faro90

Netherlands - Gasoline 98 - Constant Octane

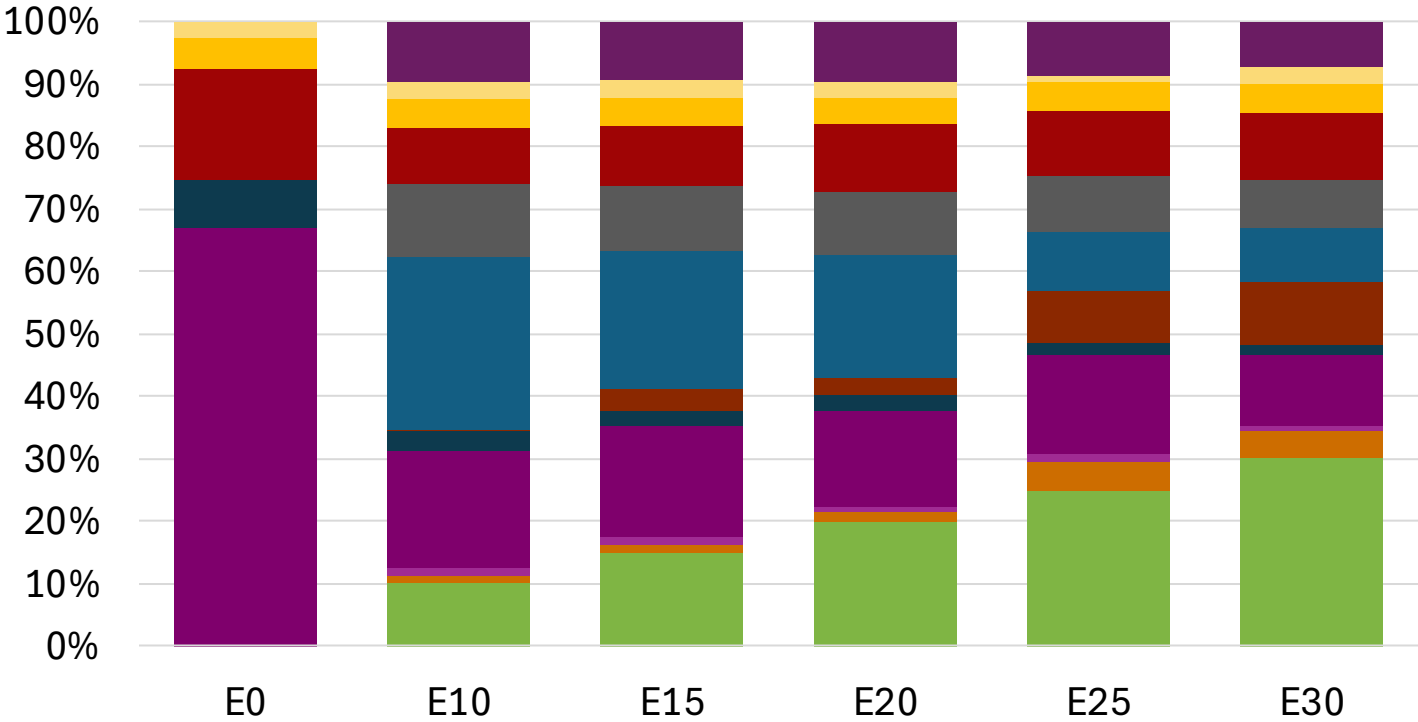
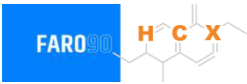


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US\$/gal)	\$ 2.409	\$ 2.290	\$ 2.216	\$ 2.164	\$ 2.057	\$ 1.994

Source: Faro90

Netherlands – Gasoline 95 – Increased Octane

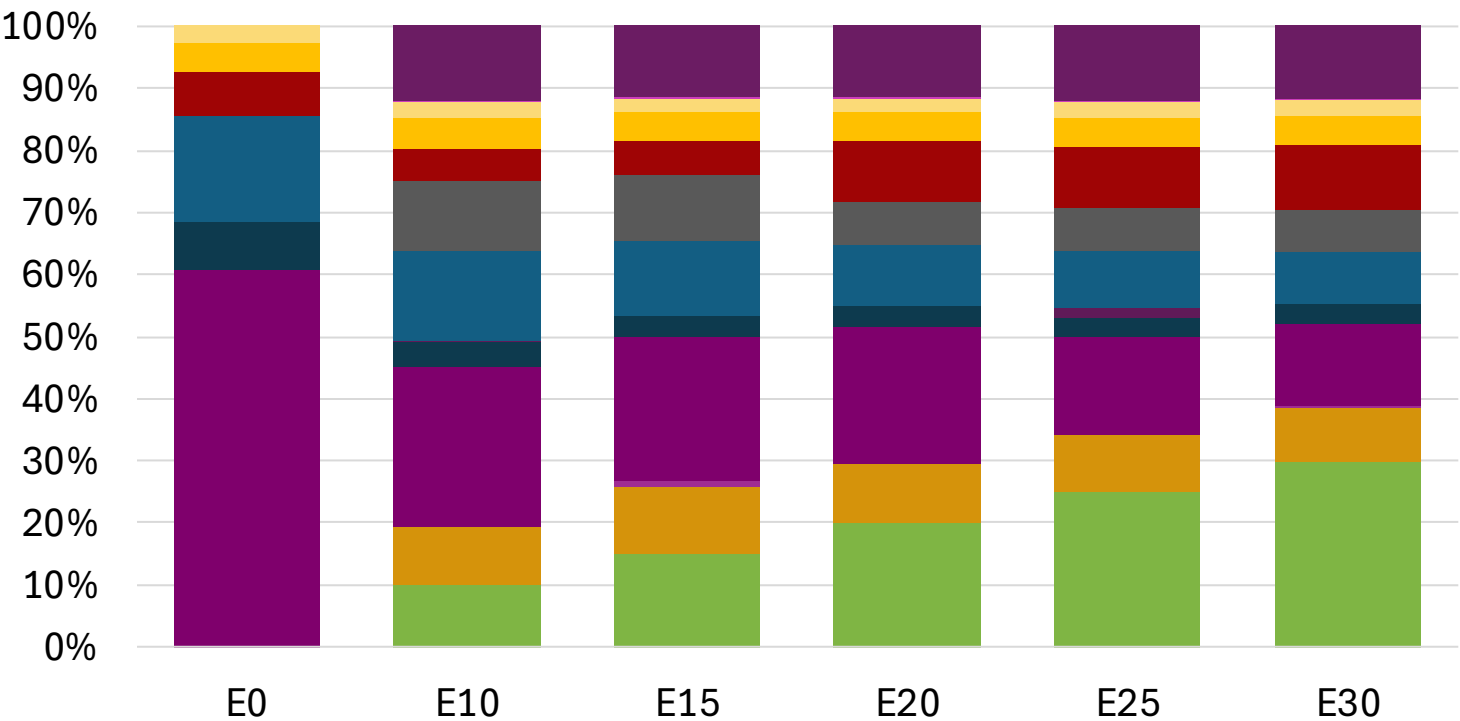
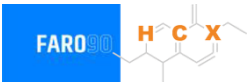


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US\$/gal)	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193

Source: Faro90

Netherlands – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US\$/gal)	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

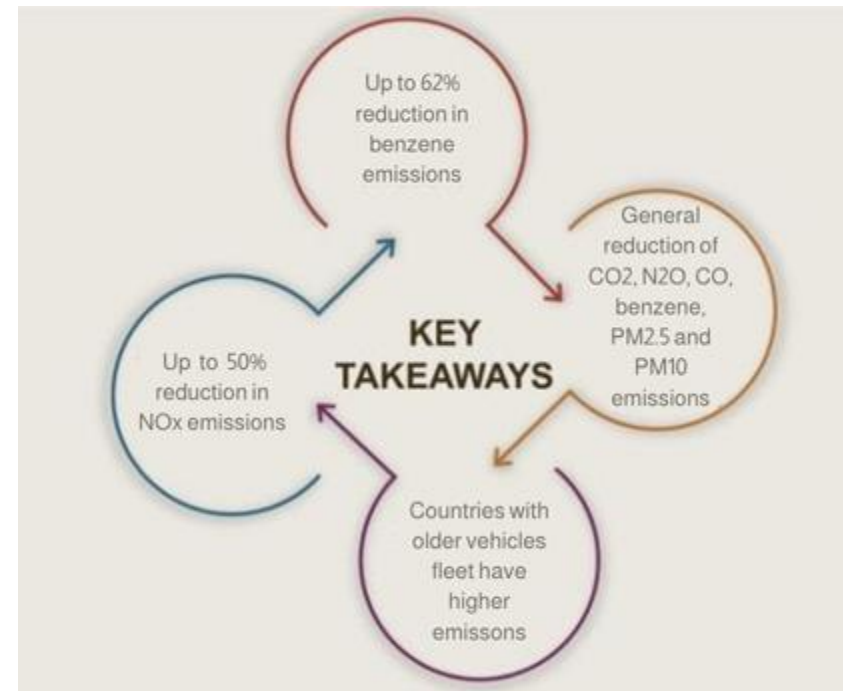
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

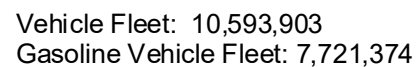
Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

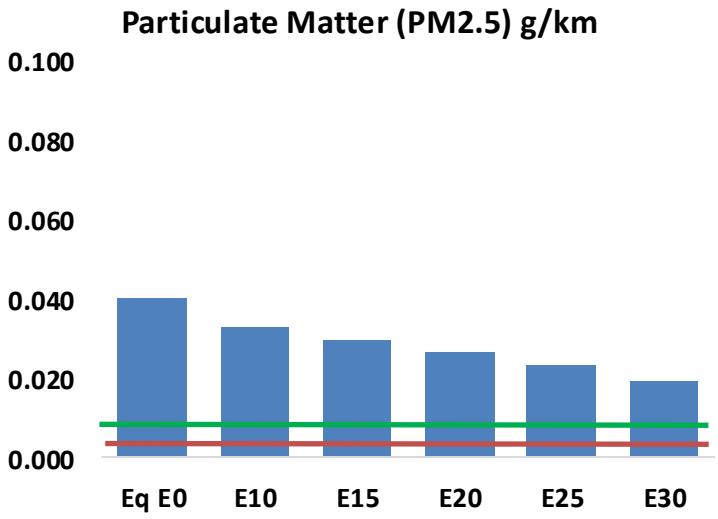
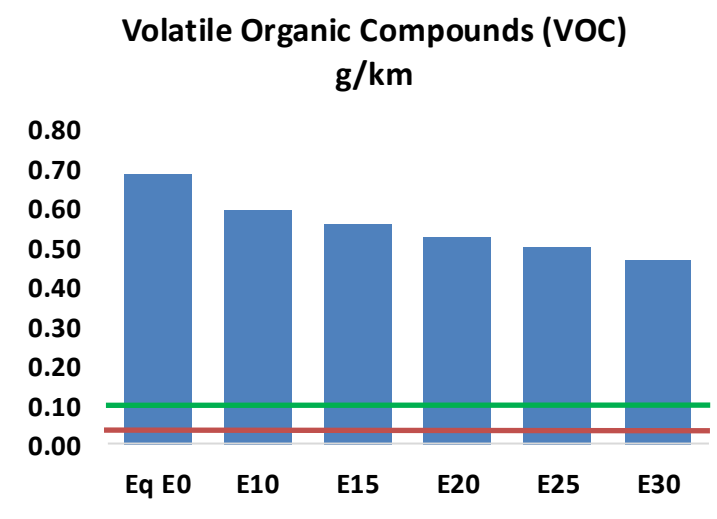
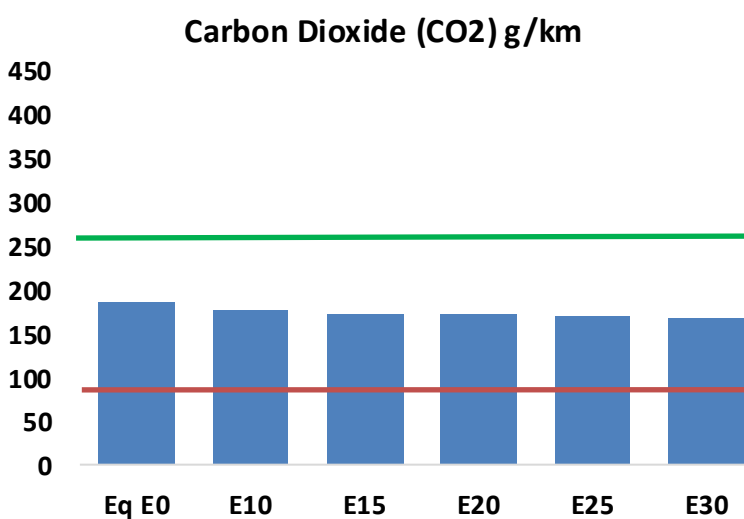
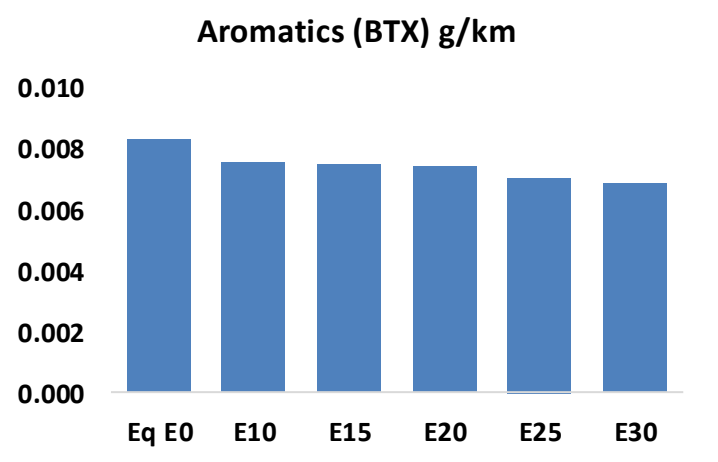
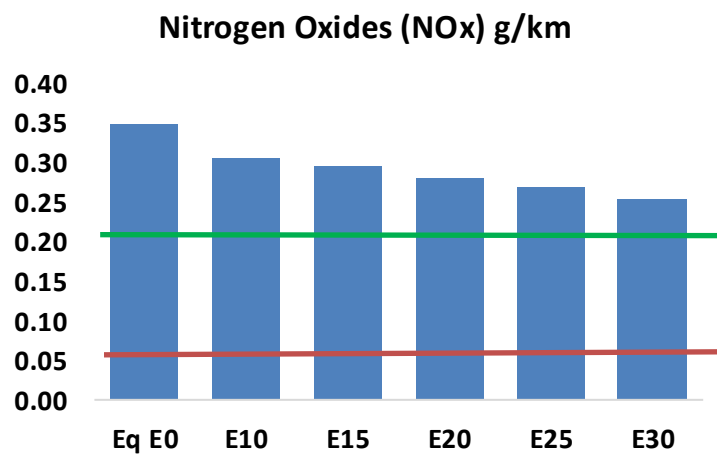
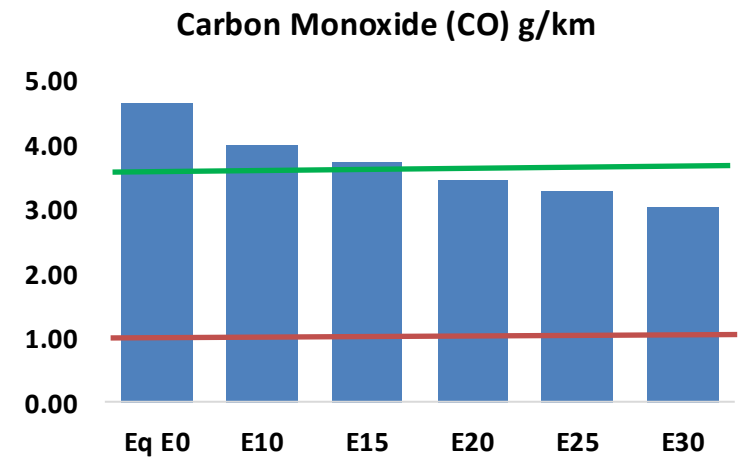
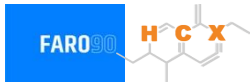
Main Results





Netherlands – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Netherlands – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	704,116	654,401	604,687	564,809	525,076	494,643	460,505	-14%	-25%
CO2	28,033,974	27,313,101	26,632,276	26,096,827	25,831,865	25,714,823	25,240,825	-5%	-8%
VOC	103,310	96,650	89,989	84,782	79,656	75,750	70,414	-13%	-23%
NOx	52,824	48,775	45,979	44,546	42,247	40,440	38,316	-13%	-20%
SOx	318	294	270	249	230	209	187	-15%	-28%
NH3	10,768	10,662	10,555	10,605	10,725	10,808	10,878	-2%	0%
N2O	470	468	466	468	478	483	489	-1%	2%
CH4	23,057	23,057	23,057	23,403	23,910	24,348	24,763	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	736	709	682	665	651	641	625	-7%	-12%
Acetaldehyde	2,046	2,348	3,445	5,246	7,148	8,168	9,651	68%	249%
Formaldehyde	7,455	7,248	8,450	9,922	10,394	11,499	12,518	13%	39%
Benzene	1,258	1,209	1,146	1,128	1,118	1,070	1,035	-9%	-11%
PM2.5	1,725	1,568	1,411	1,273	1,138	993	842	-18%	-34%
PM10	4,307	3,916	3,524	3,180	2,842	2,481	2,103	-18%	-34%

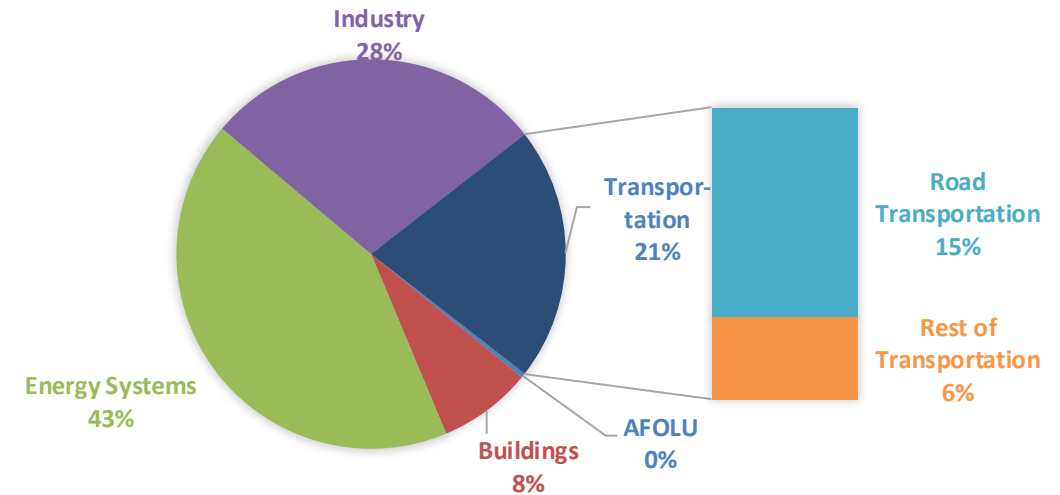
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

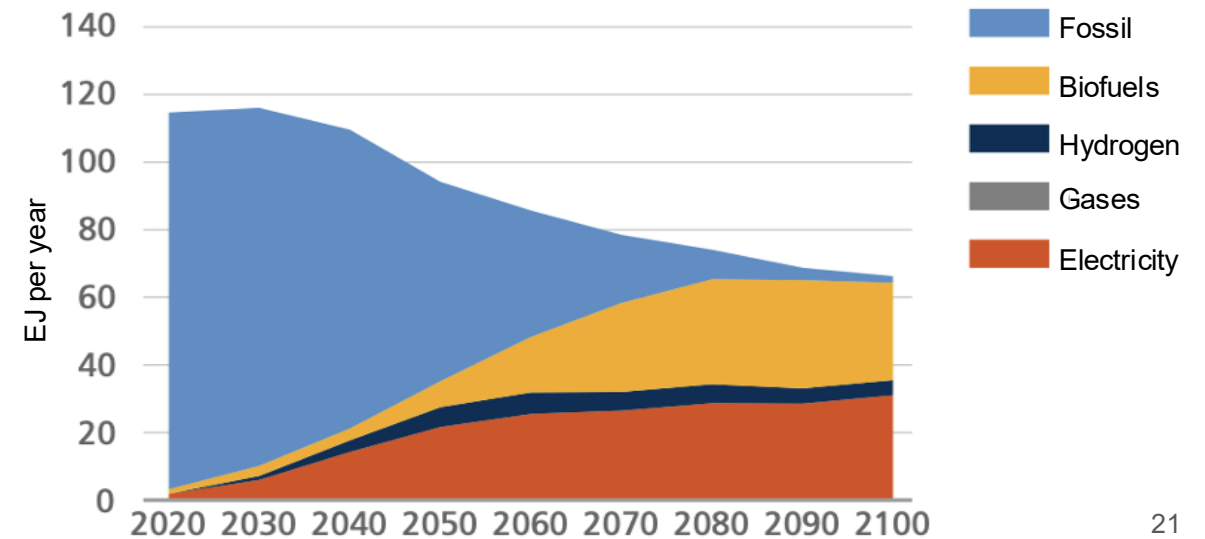
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

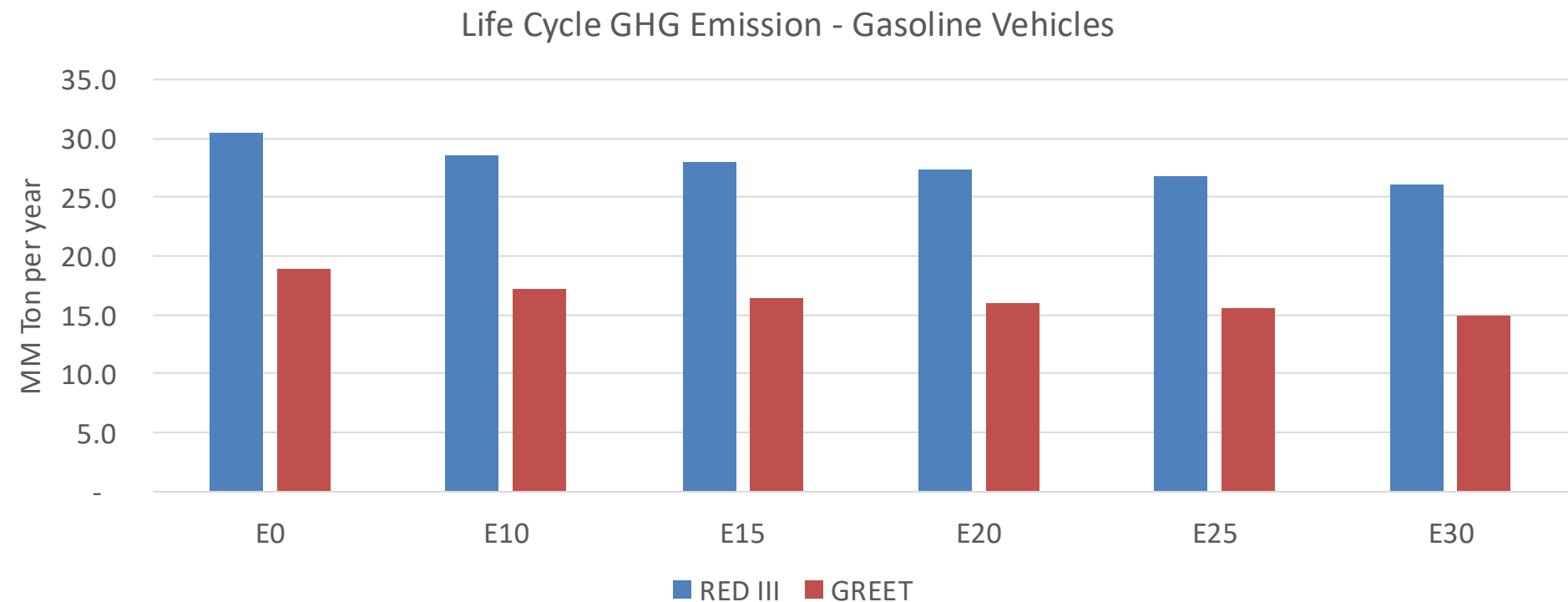
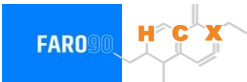
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Netherlands – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	171.1
Target @ 2035 (MMT/year)	96.3
Delta	74.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.8%	13.0%	14.8%	17.3%	20.7%
RED II/III	0.0%	6.0%	8.0%	10.1%	11.8%	14.0%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.2%	3.3%	3.7%	4.3%	5.2%
RED II/III	0.0%	2.4%	3.3%	4.1%	4.8%	5.7%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90